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AWS Networking

VPC, securing network (SG, NACL), VPC peering, S2S VPN, Direct Connect, PrivateLink, Transit Gateway

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Content:

- Amazon VPC (ENI, connectivity)
- VPC peering
- AWS site-to-site VPN
- AWS Direct Connect
- AWS VPN CloudHub
- VPC endpoints
- AWS PrivateLink
- AWS Transit Gateway
- Securing your network (RT, NACL, SG, …)
- Troubleshooting

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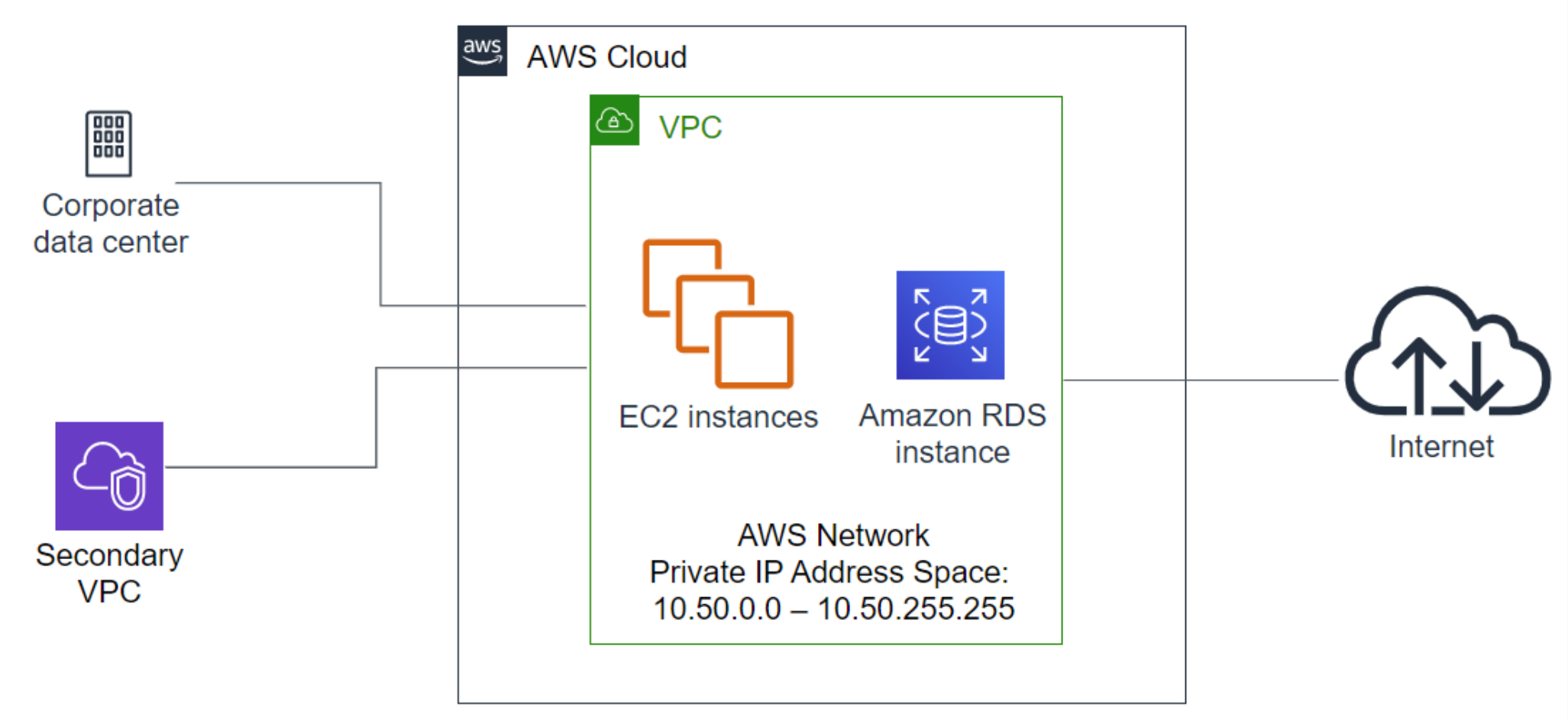
AWS Cloud networking



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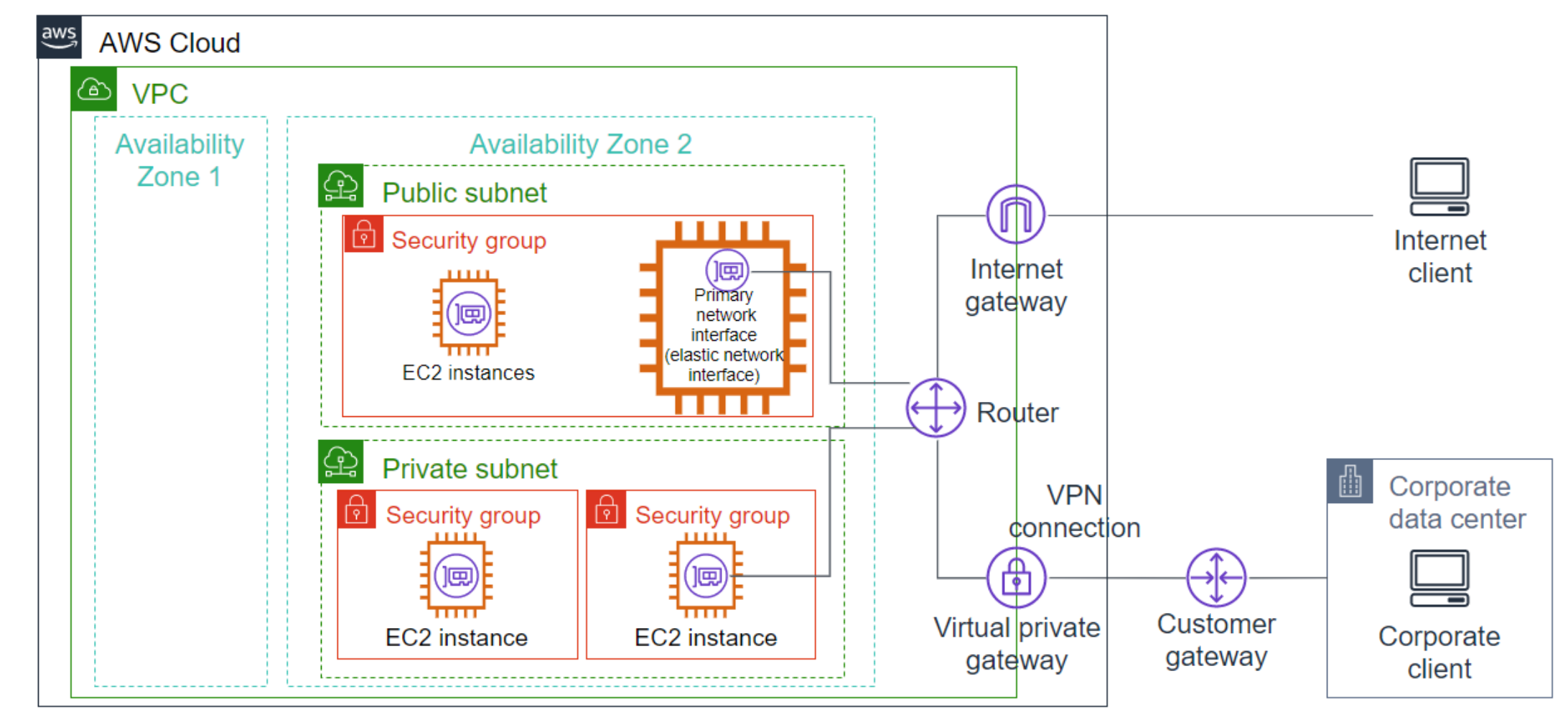
AWS networking components



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Amazon VPC

Amazon VPC

An **Amazon VPC** is a virtual network provisioned in a logically isolated section of the AWS Cloud:

- Supports logical separation with subnets
- Offers security – security groups, NACLs
- Supports VPN

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IP addressing

- Valid private IP address ranges defined by **RFC 1918**
- In an AWS VPC, you can only define networks between /16 and /28.

Best Practice

Use non-overlapping IP address ranges.

Restrict VPCs to ranges defined in RFC1918 to avoid potential routing issues.

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Amazon VPC reserved IP addresses

CIDR Block	Rationale
10.0.0.0	Network address.
10.0.0.1	Reserved by AWS for the VPC router address.
10.0.0.2	The IP address of the DNS server is always the base of the VPC network range plus two; however, VPC also reserves the base of each subnet range plus two.
10.0.0.3	Reserved by AWS for future use.
10.0.0.255	Network broadcast address. AWS does not support broadcast in a VPC, therefore this address is reserved.

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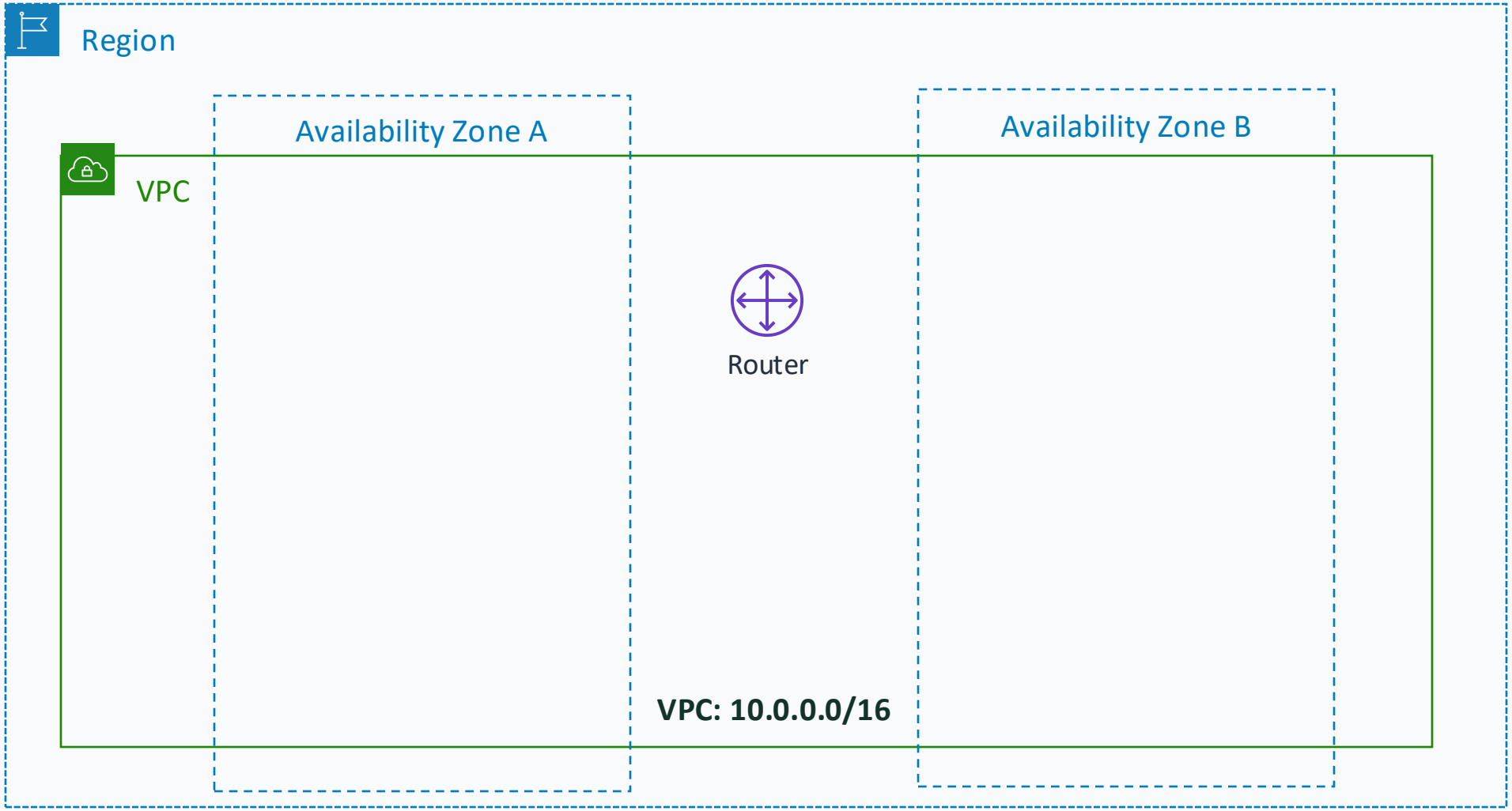
Amazon VPC characteristics



Default Route Table

Destination	Target
10.0.0.0/16	local

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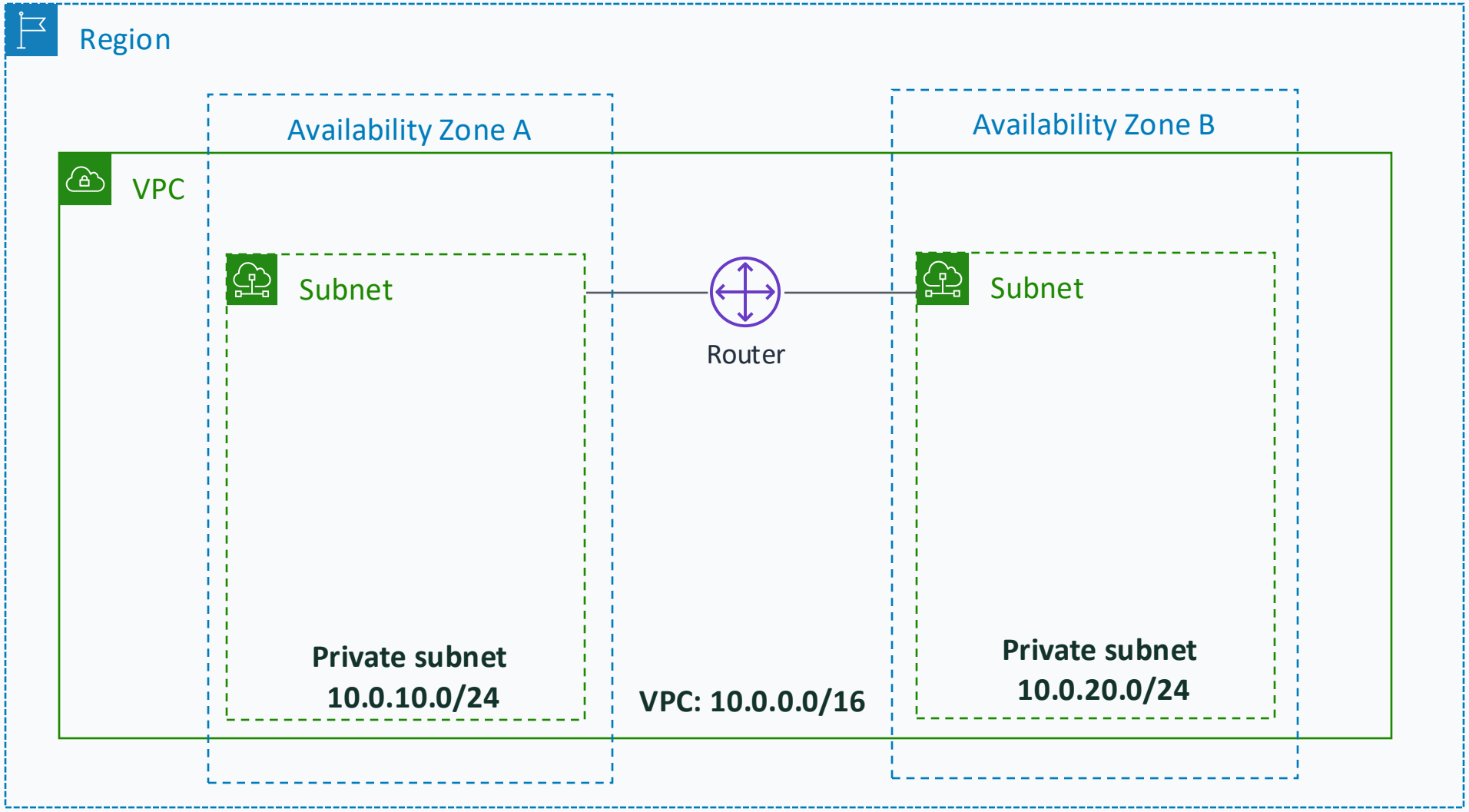
Subnets



Default Route Table

Destination	Target
10.0.0.0/16	local

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Auto-assign public IP subnet feature:

- Only instances launched in a default Amazon VPC receive a public IP address during creation.

- Instances created in custom VPCs require overriding the default setting to receive public IP addresses.

- Alternatively, configure the **Modify auto-assign IP settings** on subnets to automatically assign public IP addresses to instances.

- Unlike Elastic IP addresses, public IP addresses are not static.

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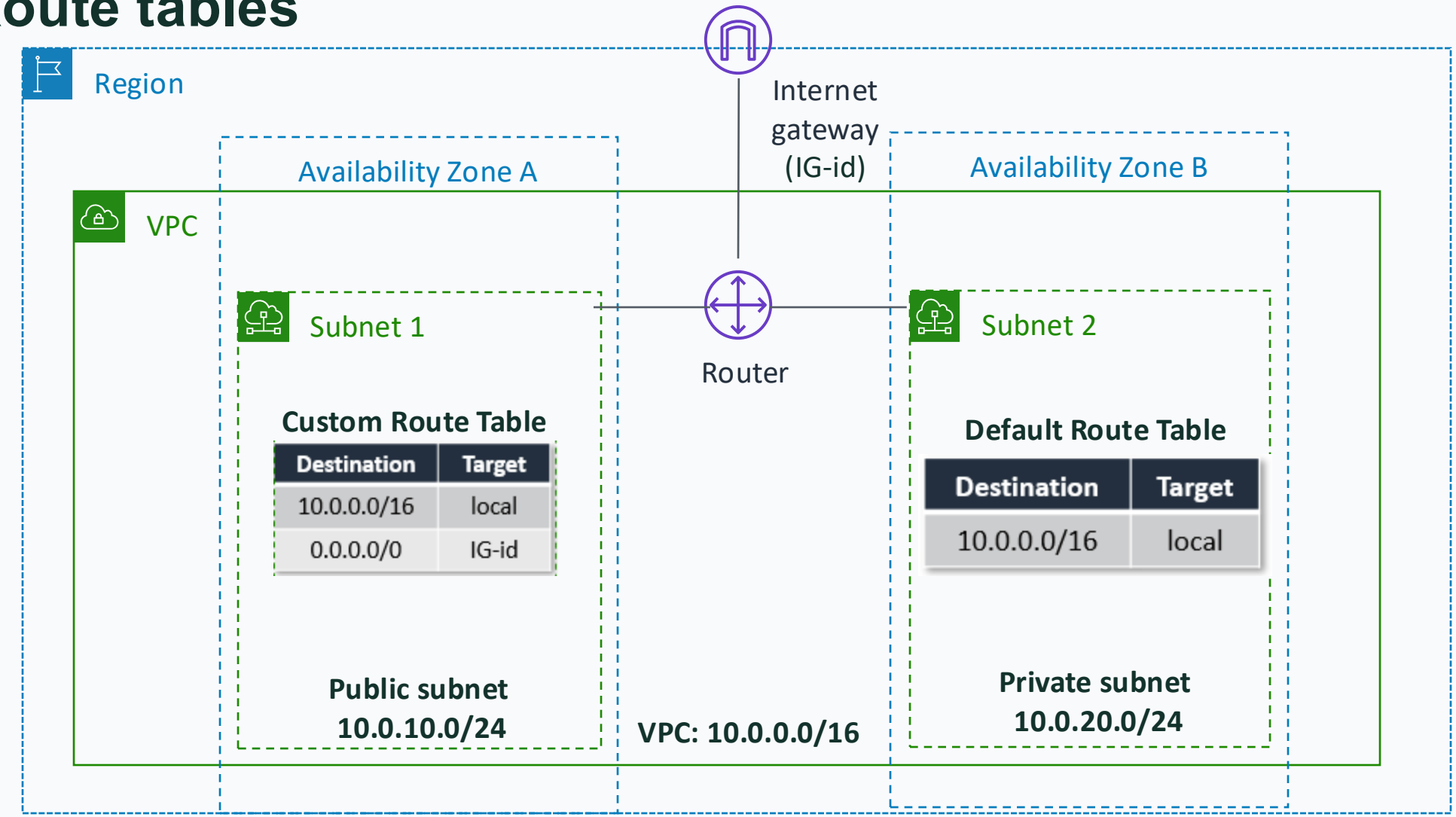
Route tables



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DNS options for a VPC

DNS server provided by Amazon (Amazon Route 53 Resolver)

Your own DNS server

Amazon Route 53 private hosted zones

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Amazon VPC connectivity options

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VPC connectivity scenarios and solutions:

If You Need to:	Consider Using:	Solution Category:
Connect a private subnet to the internet	• NAT gateway • NAT instance	• EC2 instance connectivity
Connect a VPC to another VPC	• VPC peering	• Amazon VPC to Amazon VPC
Connect a VPC to an external network	• AWS Site-to-Site VPN • AWS Direct Connect plus VPN • AWS VPN CloudHub	• Network to Amazon VPC • VPN connectivity
Connect a VPC to AWS services without leaving the AWS network	• AWS PrivateLink • VPC Gateway endpoint	• Amazon VPC to Amazon VPC • VPC Gateway endpoint
Connect a VPC to multiple VPCs and external networks	• AWS Transit Gateway	• Network to Amazon VPC • Amazon VPC to Amazon VPC

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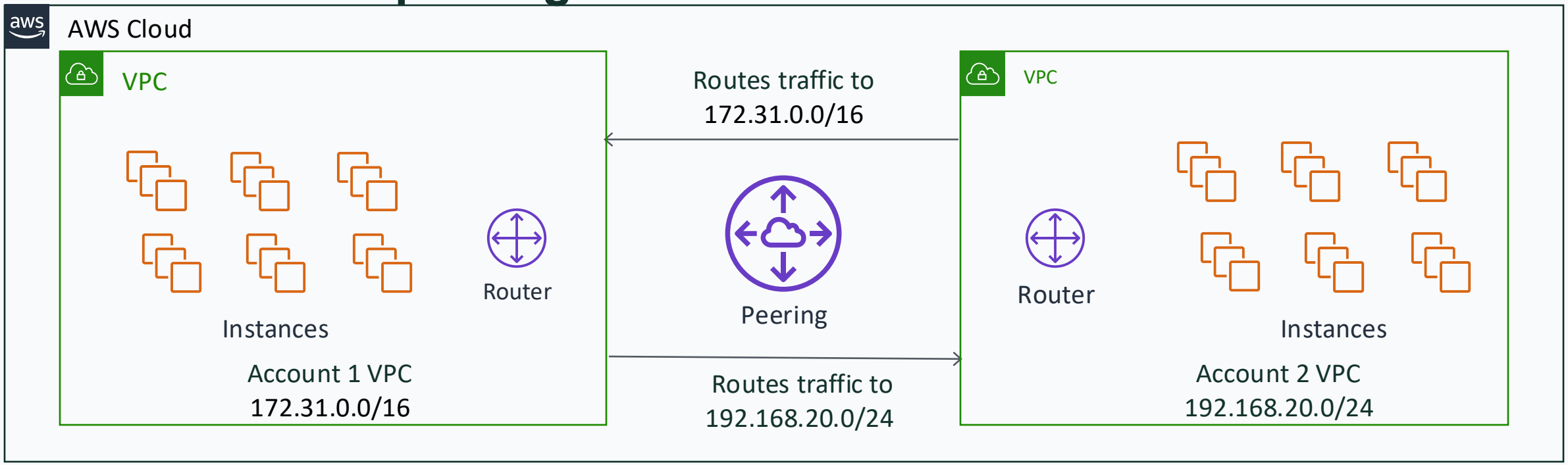
Amazon VPC peering



Private Route Table

Destination	Target
171.31.0.0/16	Local
192.168.20.0/24	VPC peer

Private Route Table

Destination	Target
192.168.20.0/24	Local
172.31.0.0/16	VPC peer

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Creating a VPC peering connection

To create a VPC peering connection:

1. Owner of requester VPC sends a VPC connection request.
2. Owner of accepter VPC accepts the VPC connection request.
3. Both owners add route table entries in both participating VPCs.
4. If necessary, owners adjust security group rules in both participating VPCs.
5. If necessary, owners enable DNS hostname resolution for VPC connection.

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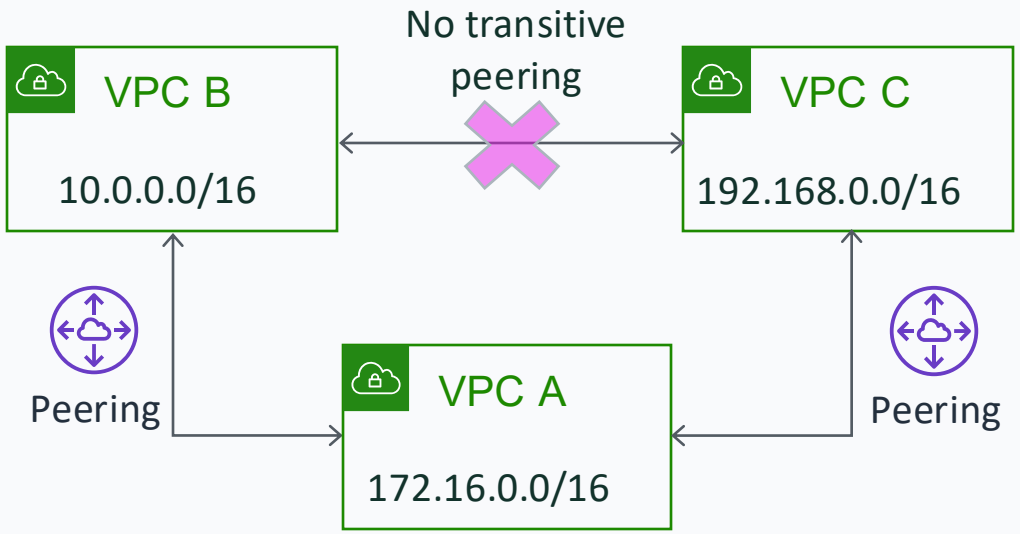
VPC peering limitations

The VPC peering limitations include:

- No overlapping IP address ranges.
- No transitive peering, edge routing, or internet gateway access.
- No NAT routing between VPCs.
- No DNS lookup resolution of private IP addresses.
- No supports cross-referencing security groups across Regions.



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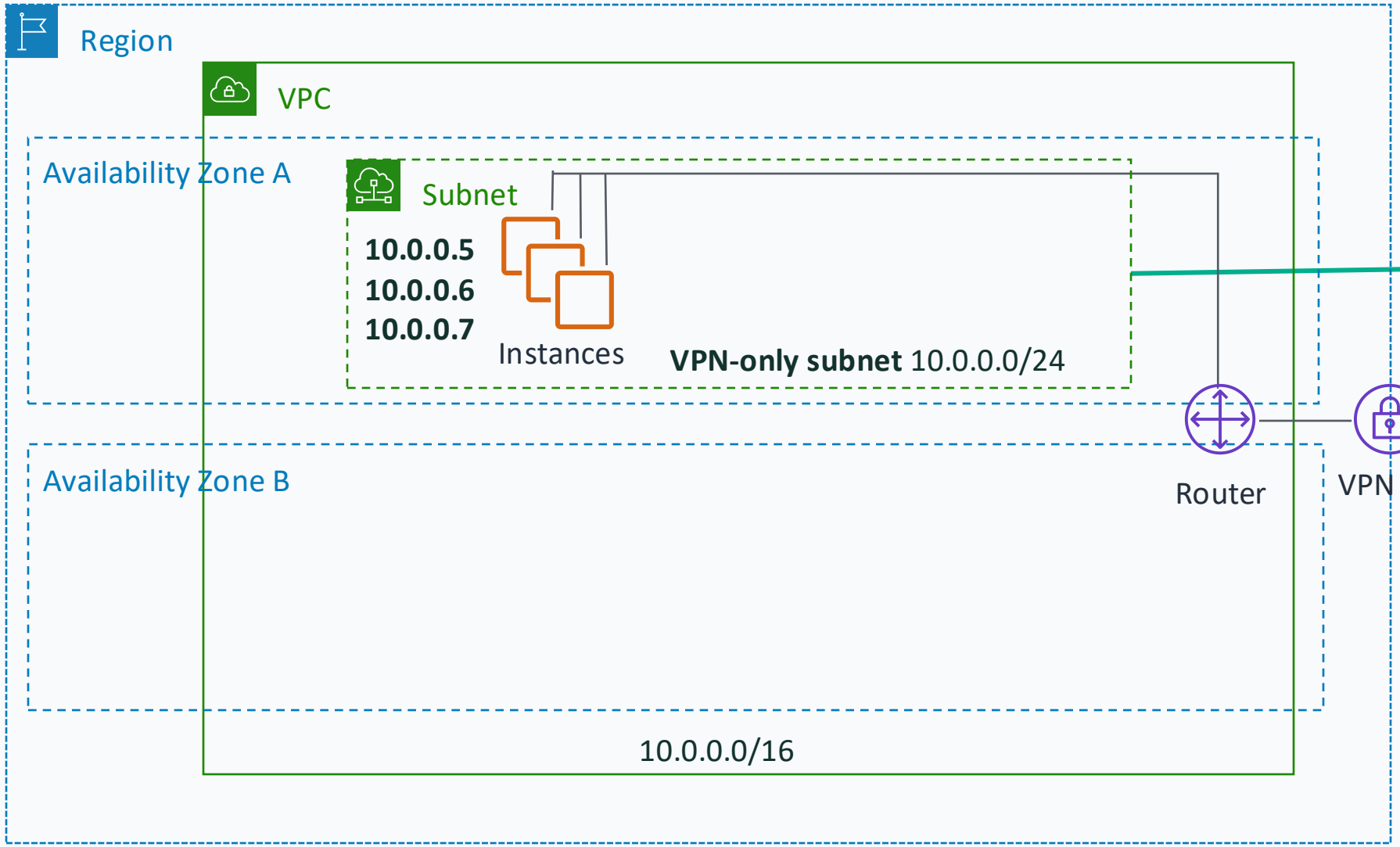
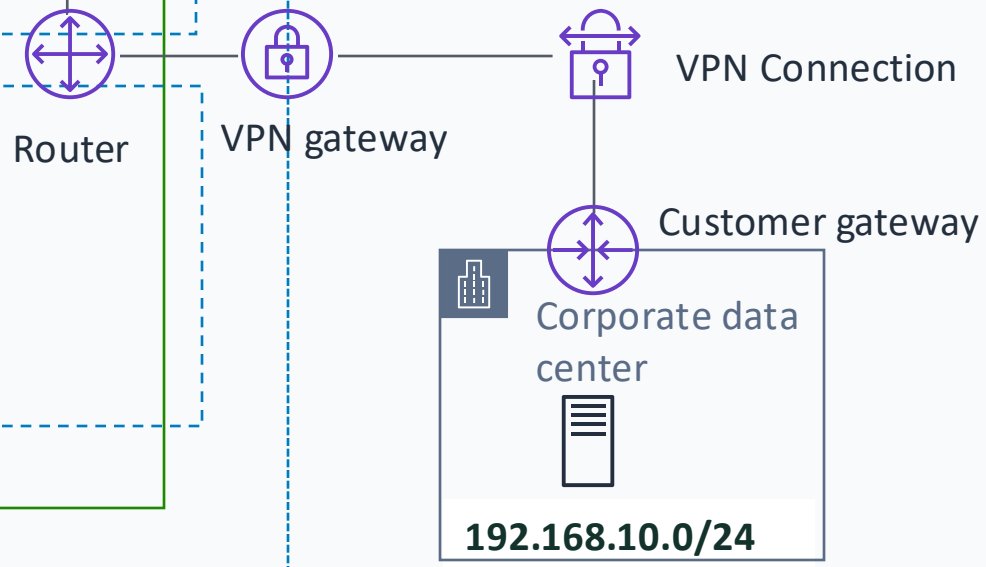
AWS site-to-site VPN



Main Route Table

Destination	Target
10.0.0.0/16	local
192.168.10.0/24	VGW-id



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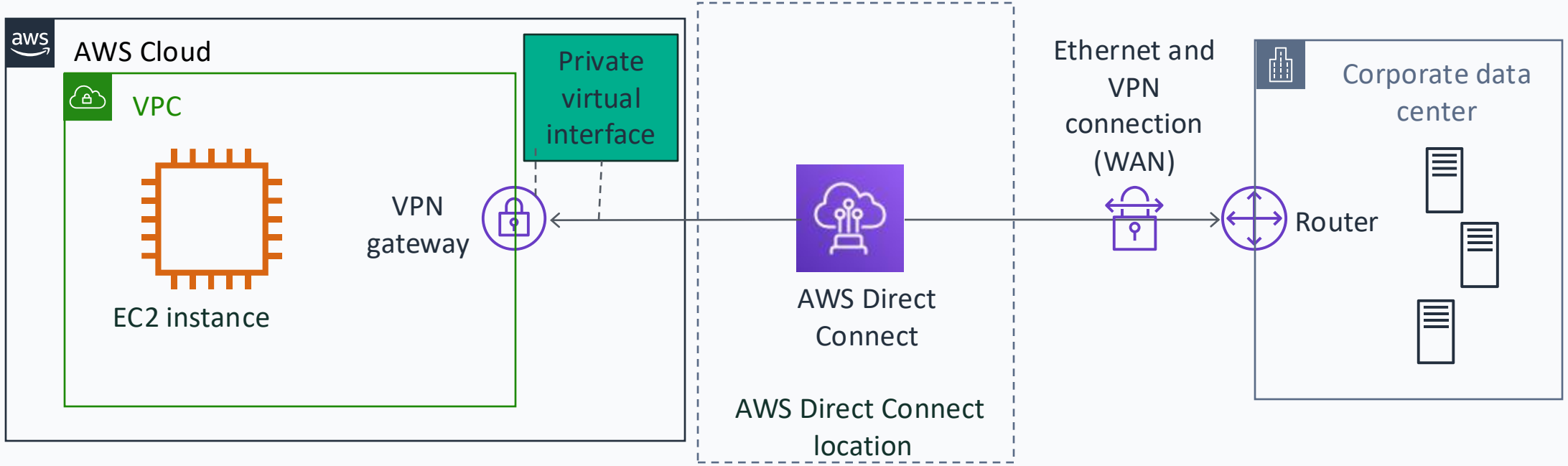
AWS Direct Connect plus VPN



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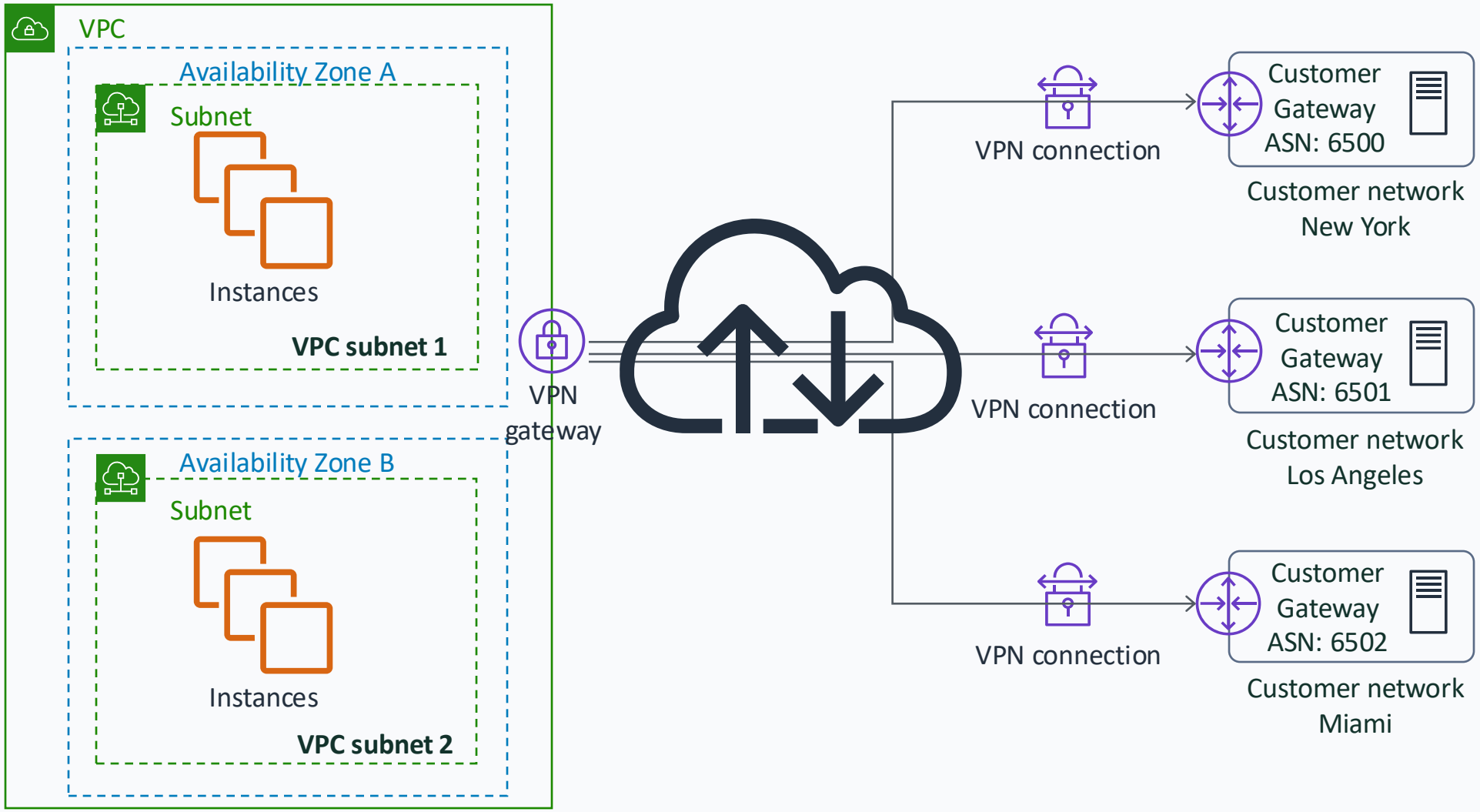
AWS VPN CloudHub



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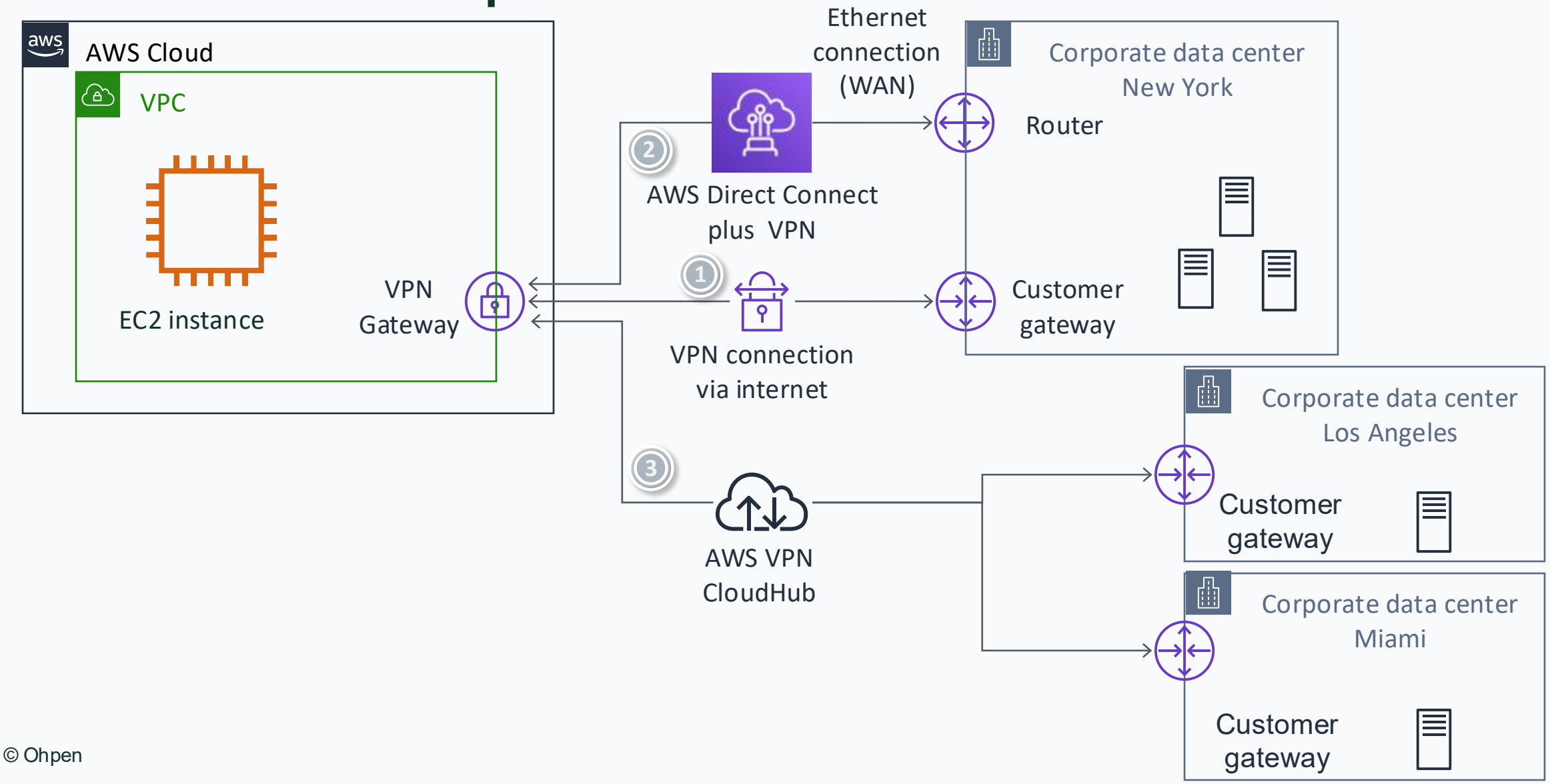
VPN connection options



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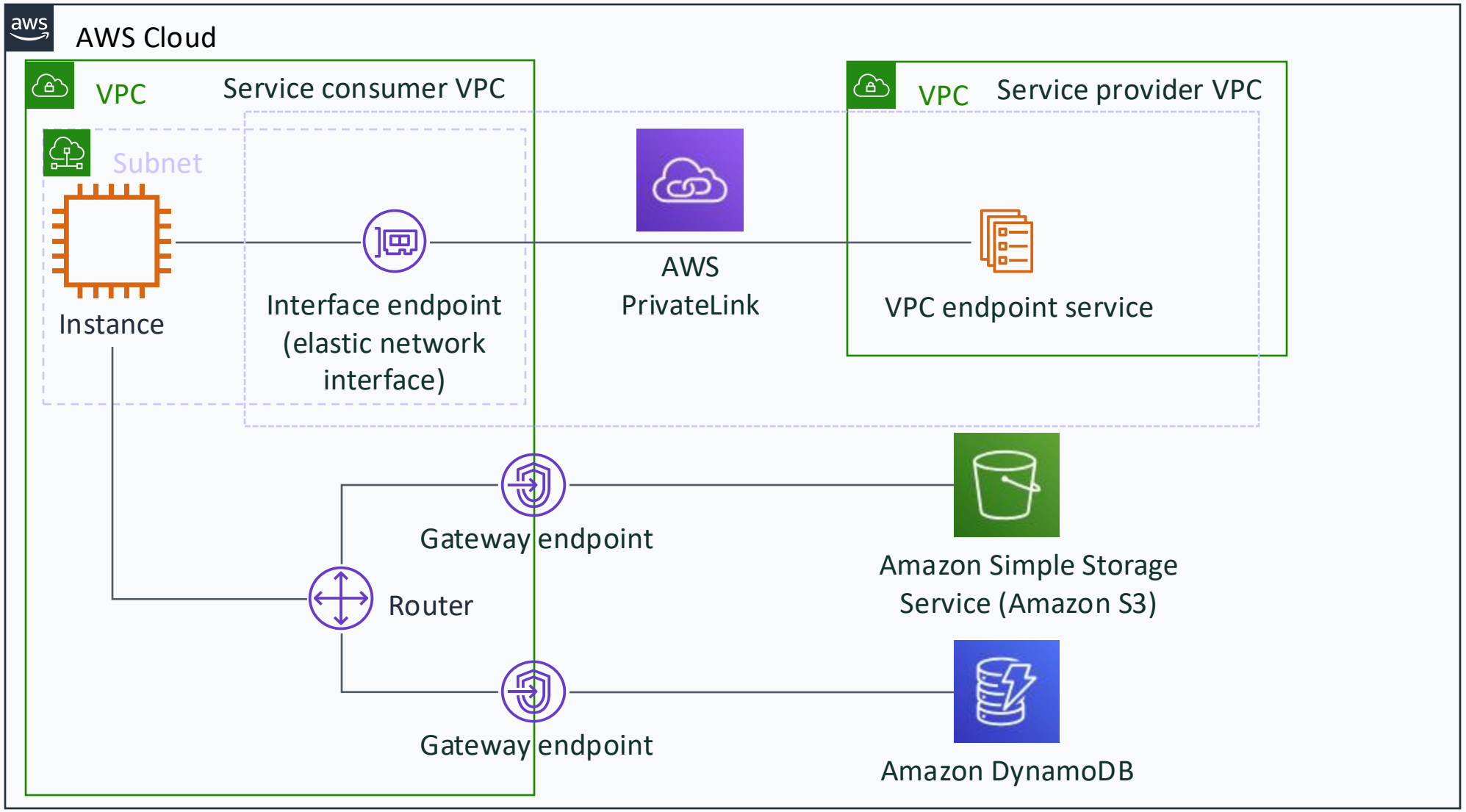
VPC endpoints



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VPC endpoints

A VPC endpoint enables you to privately connect your VPC to supported AWS services and VPC endpoint services powered by *AWS PrivateLink* without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection. Instances in your VPC do not need to know the public IP address of the resources that they want to communicate with, and the traffic between your VPC and the service provider does not leave the Amazon network.

There are two types of VPC endpoints:

•***Interface endpoint*** – An interface endpoint is a virtual component that allows you to connect to a service that is exposed through *AWS PrivateLink*. Specifically, it is an elastic network interface, which is a virtual network interface (NIC) that can be attached to an Amazon EC2 instance. This virtual card has a private IP address that serves as an entry point for traffic that is destined to the target service. Many AWS services can be powered by AWS PrivateLink, and you can also create your own application and configure it as an AWS PrivateLink-powered service. This is referred to as a *VPC endpoint service.*

•***Gateway endpoint*** – A gateway endpoint provides a mechanism to directly access resources in *Amazon S3* and *Amazon DynamoDB* without having to go through the internet. It is used as the target in a route table for a route to the resource that you are connecting to.

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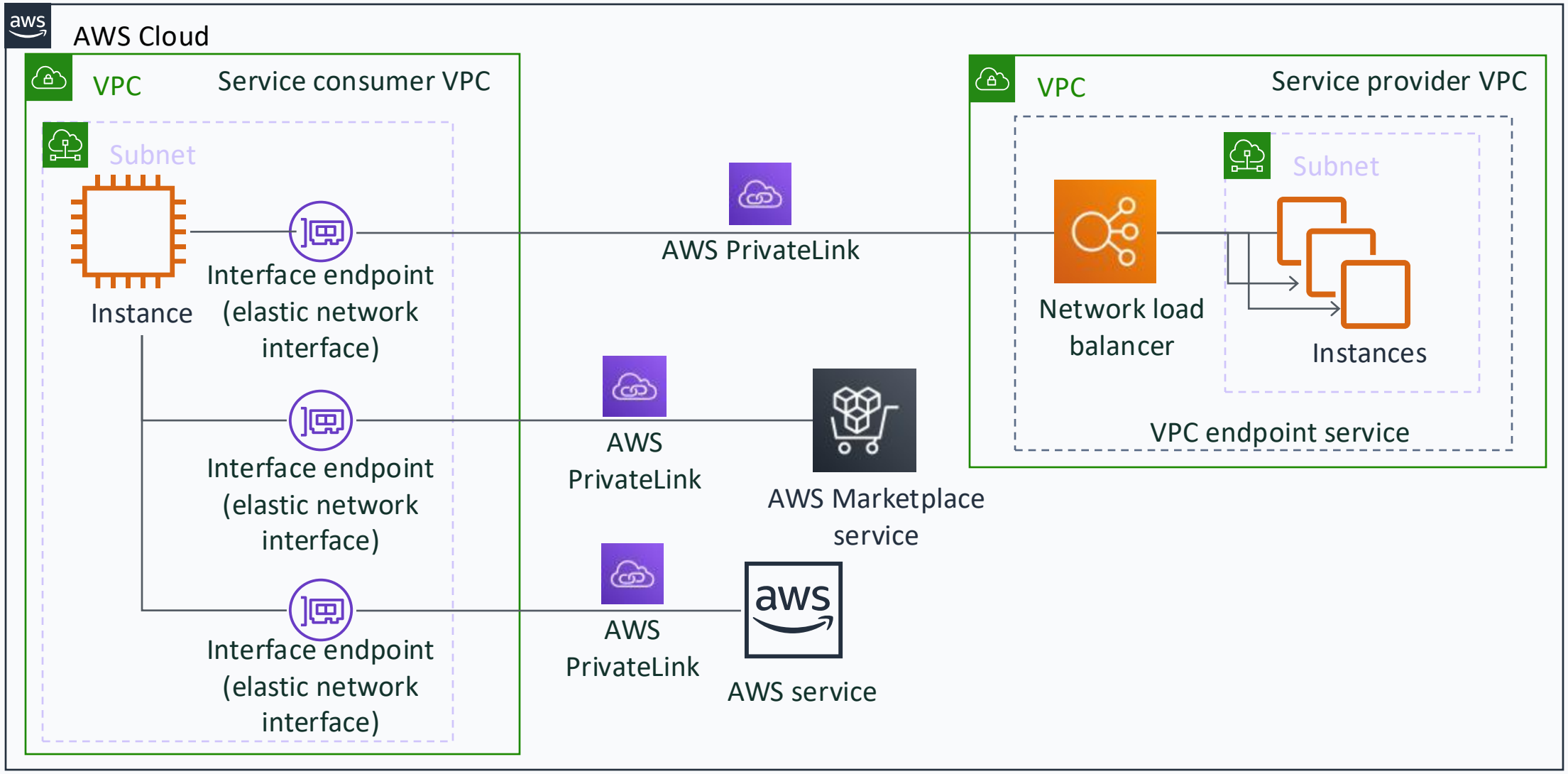
AWS PrivateLink interface endpoint



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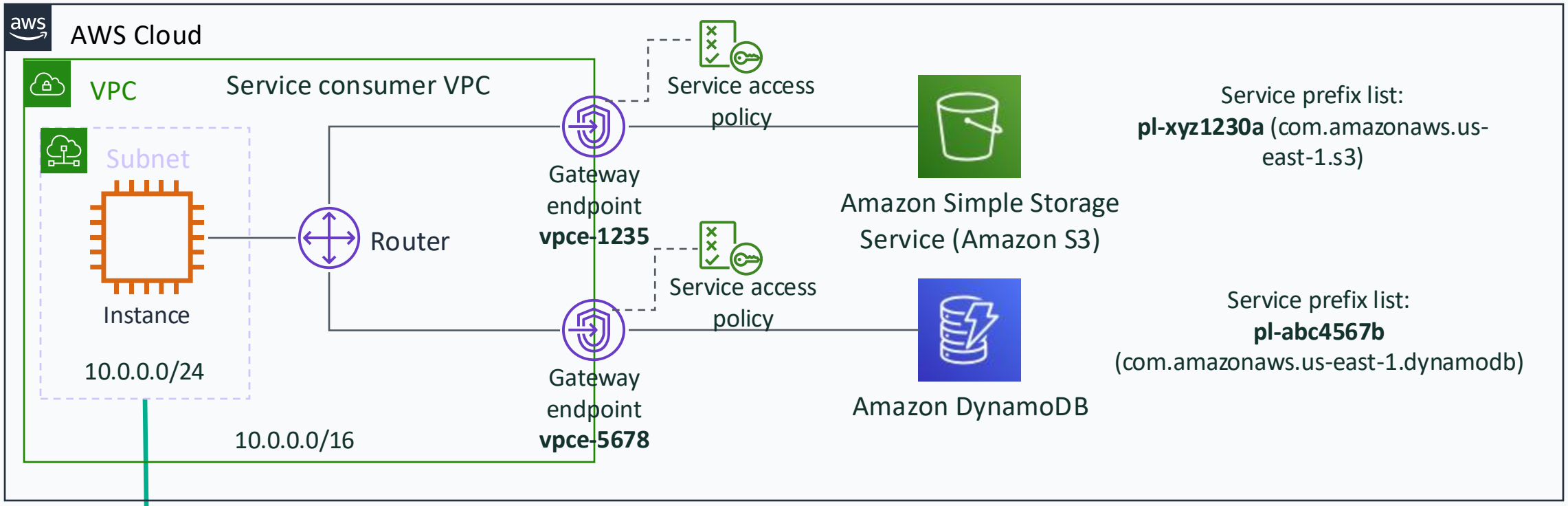
Gateway endpoint



Subnet Route Table

Destination	Target
10.0.0.0/16	Local
pl-xyz1230a	vpce-1235
pl-abc4567b	vpce-5678

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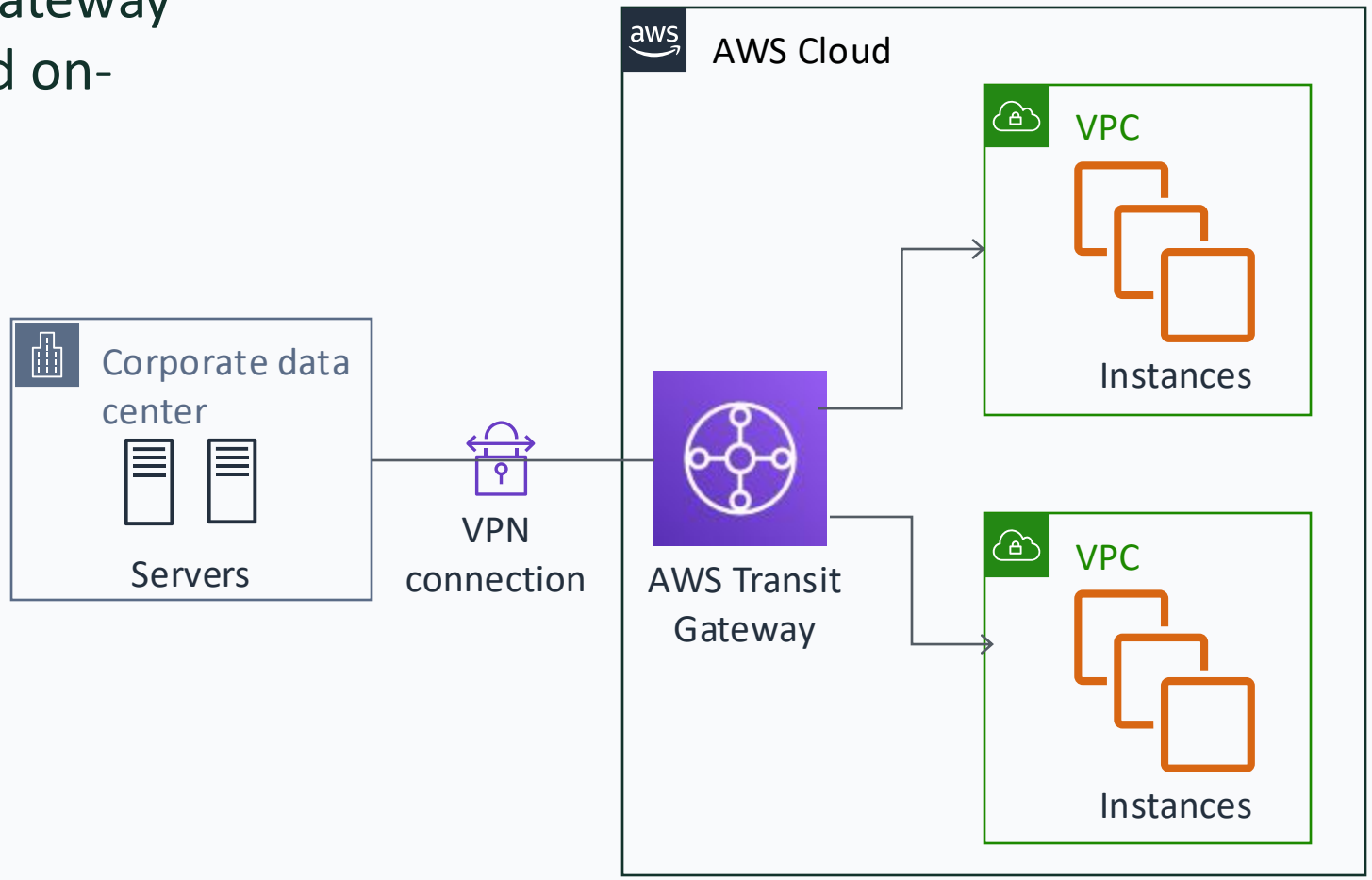
AWS Transit gateway

AWS Transit Gateway is a single gateway that connects thousands VPCs and on-premises networks:

- Attaches to VPCs and VPNs
- Supports route propagation from the route tables of the attached VPCs and VPN connections



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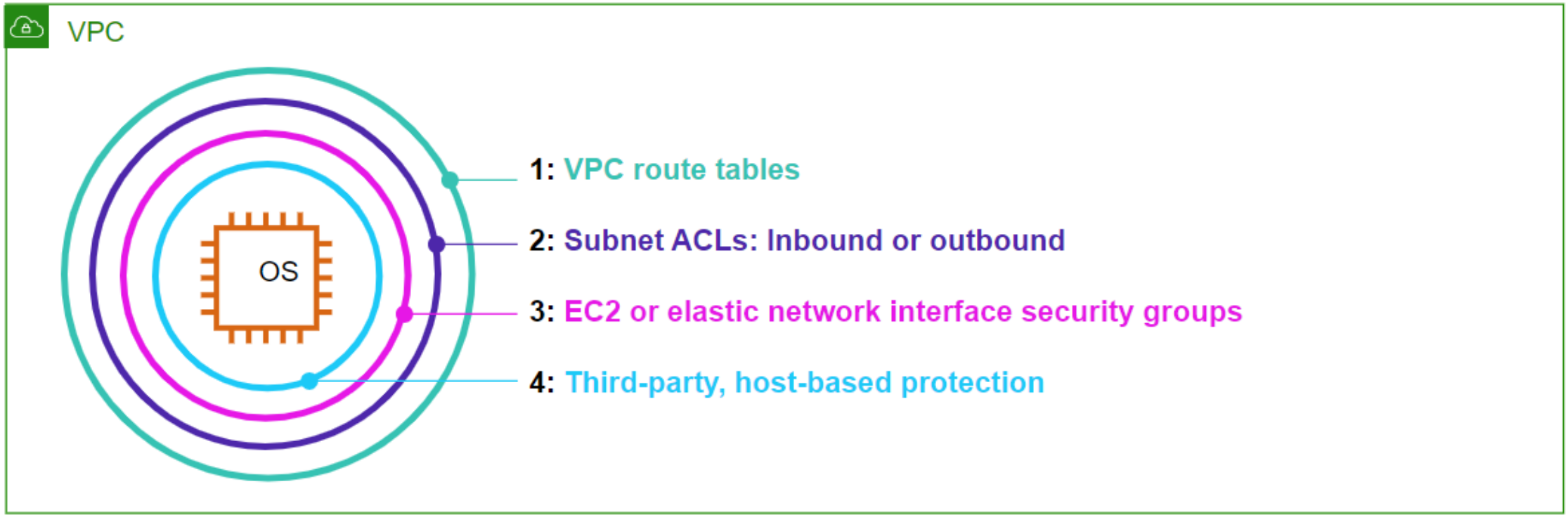
Securing your network

Layered network defense for VPCs



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Security groups

Security groups:

- Used to allow traffic to or from an elastic network interface
- Configured by default to:
  - **Deny** all inbound traffic
  - **Allow** all outbound traffic

- **Stateful**: If rules allow traffic to flow in one direction, responses can automatically flow in the opposite direction.

Outbound Rules

Type	Protocol	Port Range	Source
All traffic	All	All	0.0.0.0/0

mompopcafe security group

Inbound Rules

Type	Protocol	Port Range	Source
HTTP	TCP	80	0.0.0.0/0
SSH	TCP	22	0.0.0.0/0
MYSQL/Aurora	TCP	3306	0.0.0.0/0

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Network access control lists

- Allow or deny traffic in and out of subnets
- Default network ACL:
  - **Allows** all inbound and outbound traffic
- **Stateless**: Even if rules allow traffic to flow in one direction, you have to explicitly allow responses to flow in the opposite direction.
- Hardens security as a secondary level of defense at the subnet level

Network access control list

Inbound Rules for Default Network ACL

Rule #	Type	Protocol	Port Range	Source	Allow or Deny
100	ALL Traffic	ALL	80	0.0.0.0/0	ALLOW
*	SSH	TCP	22	0.0.0.0/0	DENY

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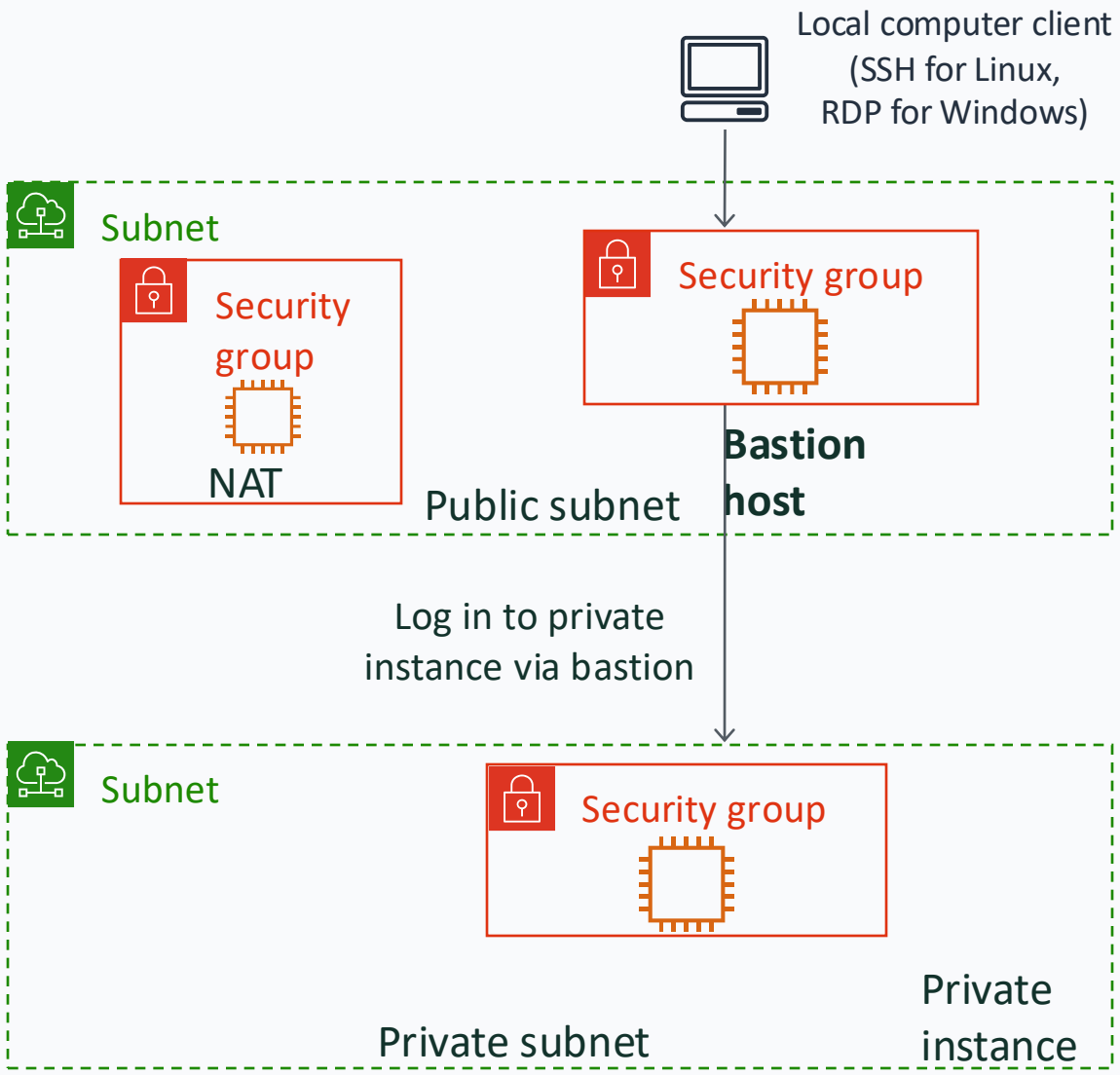
Bastion host

The **bastion host**:

- Provides public-subnet-to-private-subnet access.
  - Jump point to gain access to a private subnet
- Do not store private keys.
  - For Linux instances, use the agent forwarding feature of the SSH client to specify the key.
  - For Windows instances, decrypt password using the key in the Amazon EC2 console and then add the instance to a domain.



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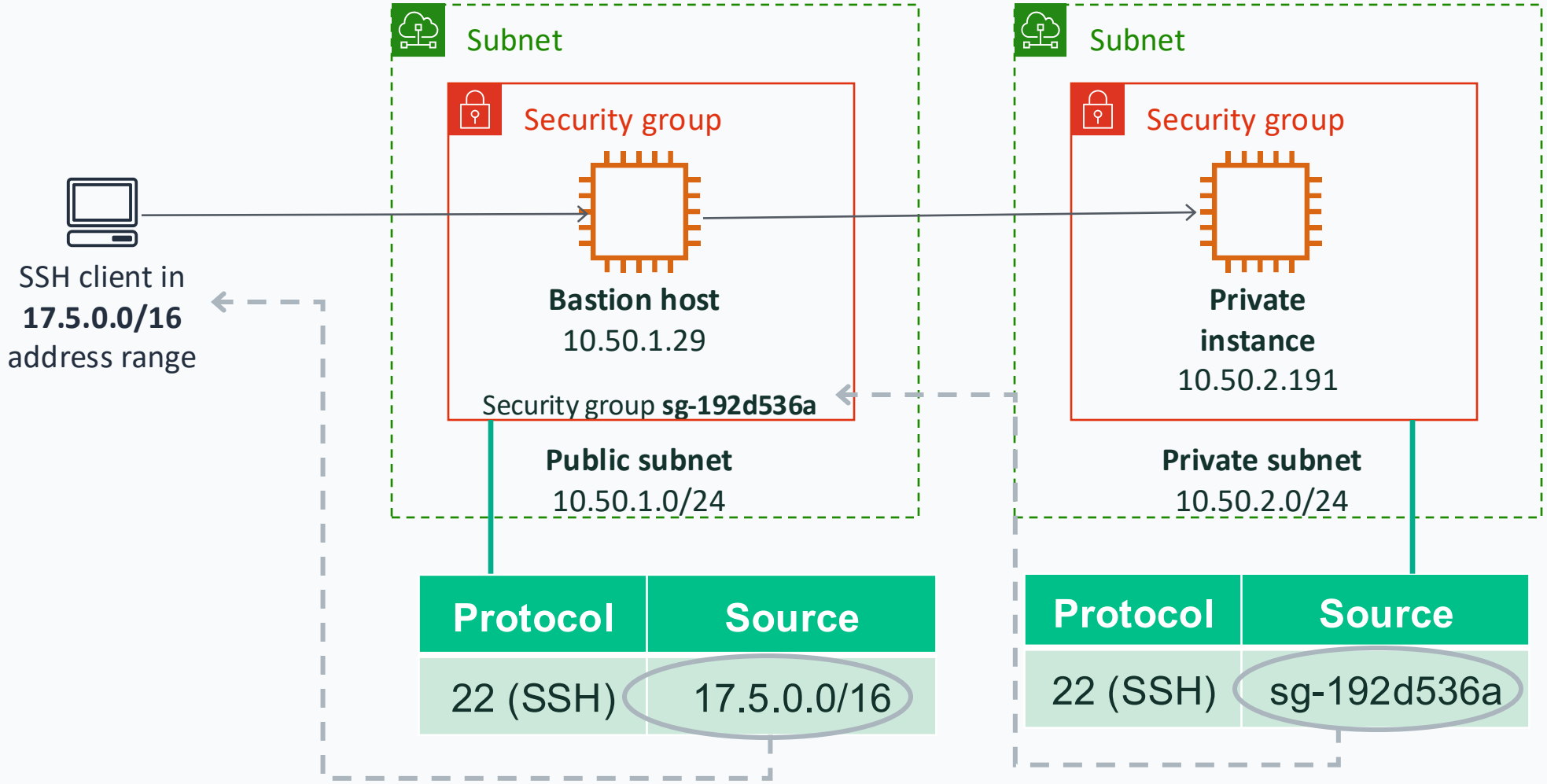
Linux bastion host security groups



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Key Takeaways

- Amazon VPC helps provision logically isolated networks on AWS.
- Subnets are classified as public or private based on their associated route tables.
- Security groups are stateful and NACLs are stateless firewalls. They are applied at elastic network interface and subnet levels respectively.

Troubleshooting networks on AWS

Common troubleshooting tasks

1. Verify that the instance is up and running.
   - Check that it has passed both the **System Status** and **Instance Status** checks.
2. Verify that the security groups associated with the instance allow connections for the required protocols and ports.
3. Verify that the network ACLs associated with the subnet allow traffic from the necessary ports and protocols.
4. Verify that the route table associated with the subnet has destination rules that point to the appropriate targets.

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Troubleshooting instance connections

Common troubleshooting tasks

- Verify that the instance is running.
- Verify security groups rules.
- Verify network ACL rules.
- Verify route table rules.

If you cannot connect to the instance via the internet:

- Verify the public IP address or DNS name you are using is correct.
- Verify the instance has a public IP address or Elastic IP address.
- Verify that an internet gateway is attached to the instance’s VPC.
- Verify that the route table of the instance’s subnet has a route rule for the destination 0.0.0.0/0 via the internet gateway.

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Troubleshooting VPC peering

Common troubleshooting tasks

- Verify that the instance is running.
- Verify security groups rules.
- Verify network ACL rules.
- Verify route table rules.

If you cannot reach resources in a peered network:

- Make sure the peering request was approved.
- **Check the security group rules:** Make sure to use the CIDR block range in VPC A to allow access from VPC B or use VPC A's security group ID.
- **Check the network ACLs:** Are you forbidding all external traffic?

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Questions?

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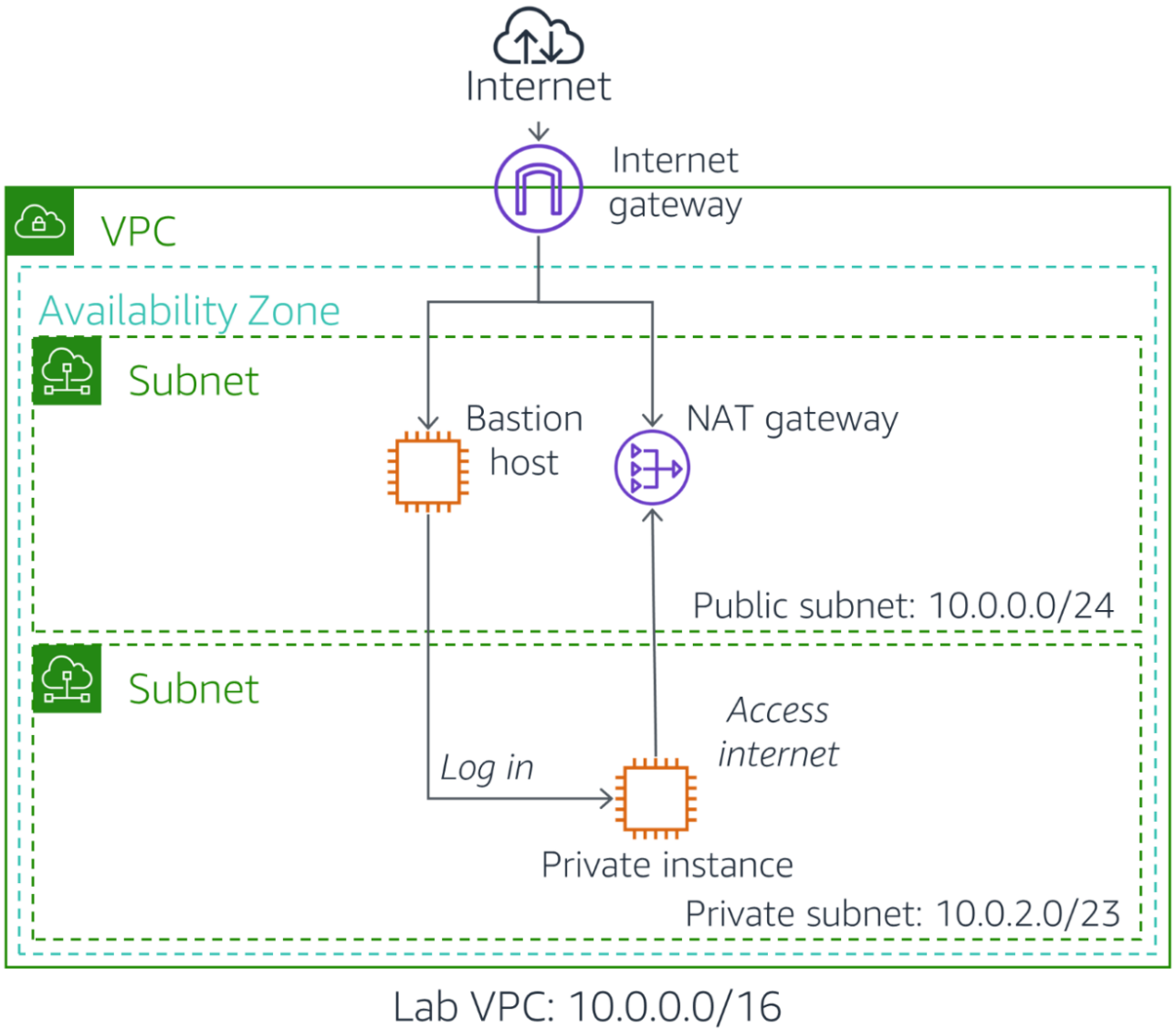
Thank you

LAB 1



Public Route Table

Destination	Target
10.0.0.0/16	Local
0.0.0.0/0	Internet gateway

Private Route Table

Destination	Target
10.0.0.0/16	Local
0.0.0.0/0	NAT gateway

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1. Create bastion host with bastion ssh key
    1. Allow inbound traffic from your public IP for port 22
2. Create instance in private subnet -> with own ssh key (do not use the same key as for bastion instance)
    1. Allow inbound traffic only from VPC CIDR for port 22
3. Use basic-network CF for networking
4. Connect to the private instance via the bastion host (SSH)

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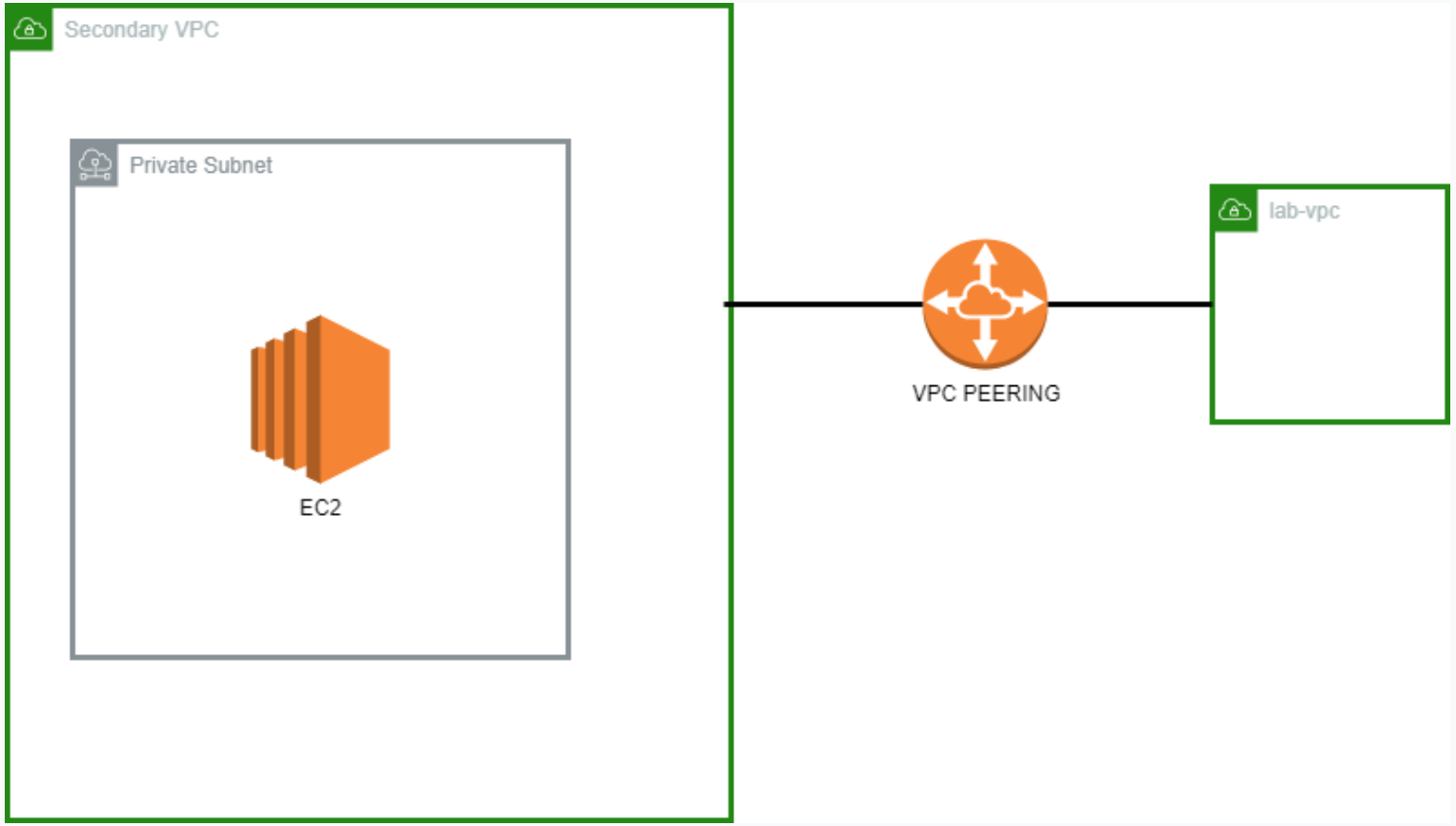
Secondary VPC with vpc peering connection to lab-vpc



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1. Create secondary VPC with one private subnet
2. Deploy static EC2 instance into the private subnet
3. Create VPC peering connection between secondary VPC and lab-vpc
4. Set up routing and security rules
5. Test the connection from the private instance in the lab-vpc to the private instance in the secondary VPC (ping, ssh …)

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